

HDOC Day-11

Question: Menu-Driven Program for Digit Operations Using Loops

1. Problem Statement & Identification

- **Problem:** Create a menu-driven program using switch-case to input a positive integer and calculate either:
 1. Sum of squares of digits using a while loop.
 2. Sum of cubes of digits using a for loop.
 3. Difference between sum of cubes and sum of squares using a do-while loop.
- **Input:** A positive integer (num) and menu choice (choice).
- **Output:** Calculated result based on the chosen operation.

2. Algorithm

1. **Start**
2. Input a positive integer num.
3. Display menu options (1 to 3).
4. Read user choice.

5. Execute operations based on choice using switch-case:

- **Case 1:** Set temp = num. While temp > 0, extract digit (temp % 10), add digit * digit to sum_squares, and update temp = temp / 10. Print sum_squares.
- **Case 2:** For temp = num, condition temp > 0, step temp /= 10, extract digit, add digit * digit * digit to sum_cubes. Print sum_cubes.
- **Case 3:** Set temp = num. Using a do-while loop, extract digit, compute both sum_squares and sum_cubes, and update temp = temp / 10 until temp == 0. Calculate diff = sum_cubes - sum_squares. Print diff.
- **Default:** Print "Invalid Choice!".

6. End

3. Test Case Table

Test Case	Input Number	Choice	Operation & Loop	Expected Output	Actual Output	Status
1	123	1	Sum of square	14	14	Pass

Tes t Ca se	Input Num ber	Choi ce	Operat ion & Loop	Expec ted Outpu t	Actu al Out put	Stat us
			s (while)			
2	123	2	Sum of cubes (for)	36	36	Pass
3	123	3	Differe nce (do- while)	22	22	Pass
4	45	1	Sum of square s (while)	41	41	Pass

4. C Source Code

```
#include <stdio.h>
```

```
int main(void) {
```

```
    int choice, num, temp, digit;
```

```
    int sum_squares = 0, sum_cubes = 0, diff = 0;
```

```

printf("Enter a positive integer: ");
if (scanf("%d", &num) != 1 || num <= 0) {
    printf("Invalid input! Please enter a positive integer.\n");
    return 1;
}

printf("\n--- MENU ---\n");
printf("1. Sum of squares of digits (using while loop)\n");
printf("2. Sum of cubes of digits (using for loop)\n");
printf("3. Difference between sum of cubes and sum of squares (using
do-while loop)\n");
printf("Enter your choice (1-3): ");
scanf("%d", &choice);

switch (choice) {
    case 1:
        temp = num;
        while (temp > 0) {
            digit = temp % 10;
            sum_squares += (digit * digit);
            temp /= 10;
        }
        printf("\nNumber: %d, Sum of squares of digits = %d\n", num,
sum_squares);
        break;

    case 2:
        for (temp = num; temp > 0; temp /= 10) {
            digit = temp % 10;
            sum_cubes += (digit * digit * digit);
        }
        printf("\nNumber: %d, Sum of cubes of digits = %d\n", num,
sum_cubes);
        break;

    case 3:
        temp = num;
        sum_squares = 0;
        sum_cubes = 0;
        do {

```

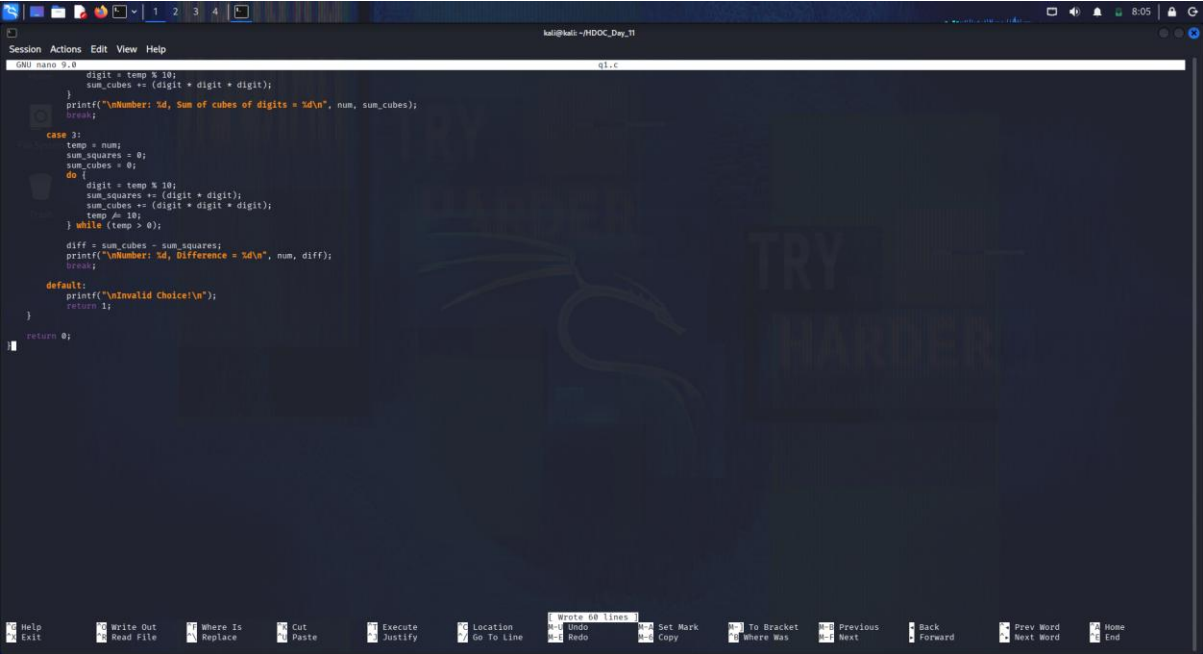
```
        digit = temp % 10;
        sum_squares += (digit * digit);
        sum_cubes += (digit * digit * digit);
        temp /= 10;
    } while (temp > 0);

    diff = sum_cubes - sum_squares;
    printf("\nNumber: %d, Difference = %d\n", num, diff);
    break;

default:
    printf("\nInvalid Choice!\n");
    return 1;
}

return 0;
}
```

5. Compilation, Execution & Screenshots



```
kali@kali: ~/HDOC_Day_11
Session Actions Edit View Help
(kali@kali)-[~]
$ mkdir HDOC_Day_11
cd HDOC_Day_11
(kali@kali)-[~/HDOC_Day_11]
$ nano q1.c
(kali@kali)-[~/HDOC_Day_11]
$ gcc q1.c -o q1
./q1
Enter a positive integer: 123
— MENU —
1. Sum of squares of digits (using while loop)
2. Sum of cubes of digits (using for loop)
3. Difference between sum of cubes and sum of squares (using do-while loop)
Enter your choice (1-3): 1
Number: 123, Sum of squares of digits = 14
```